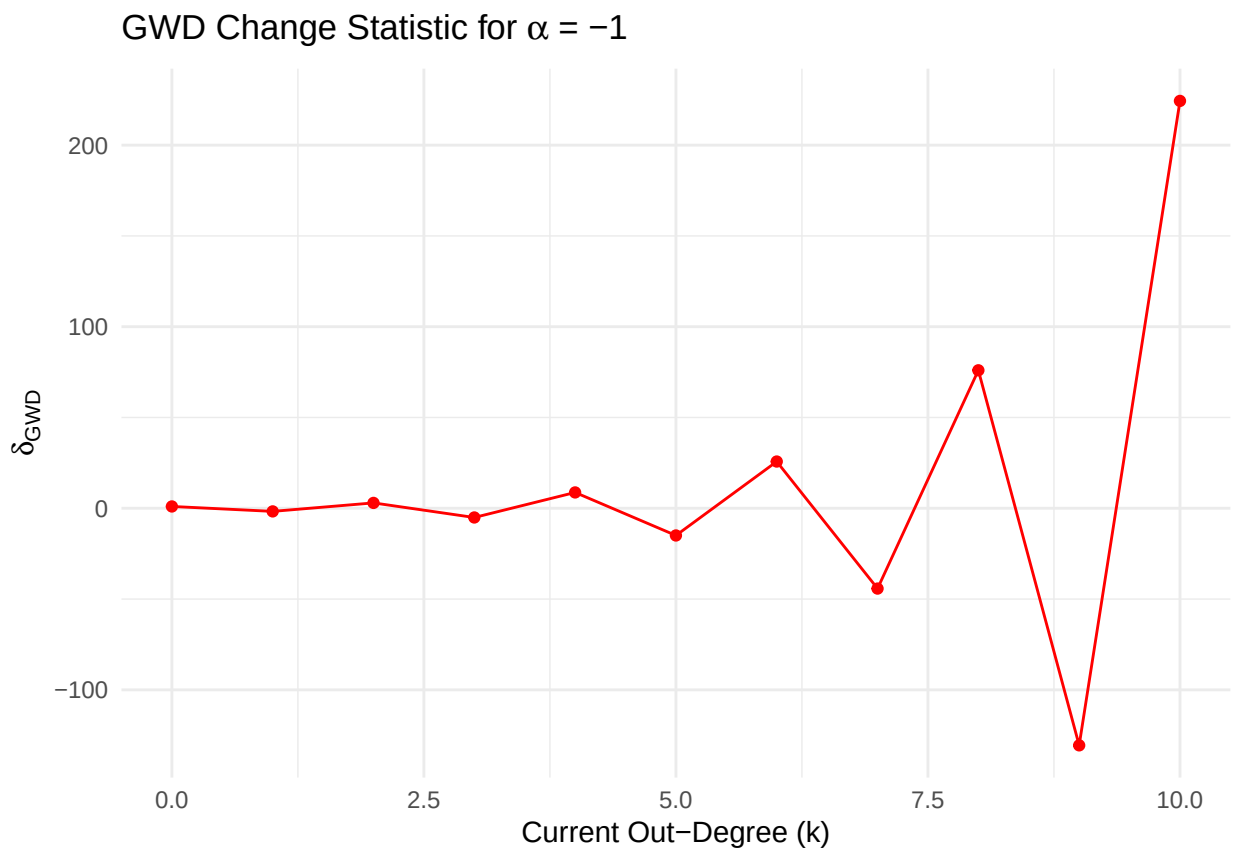
Task 1

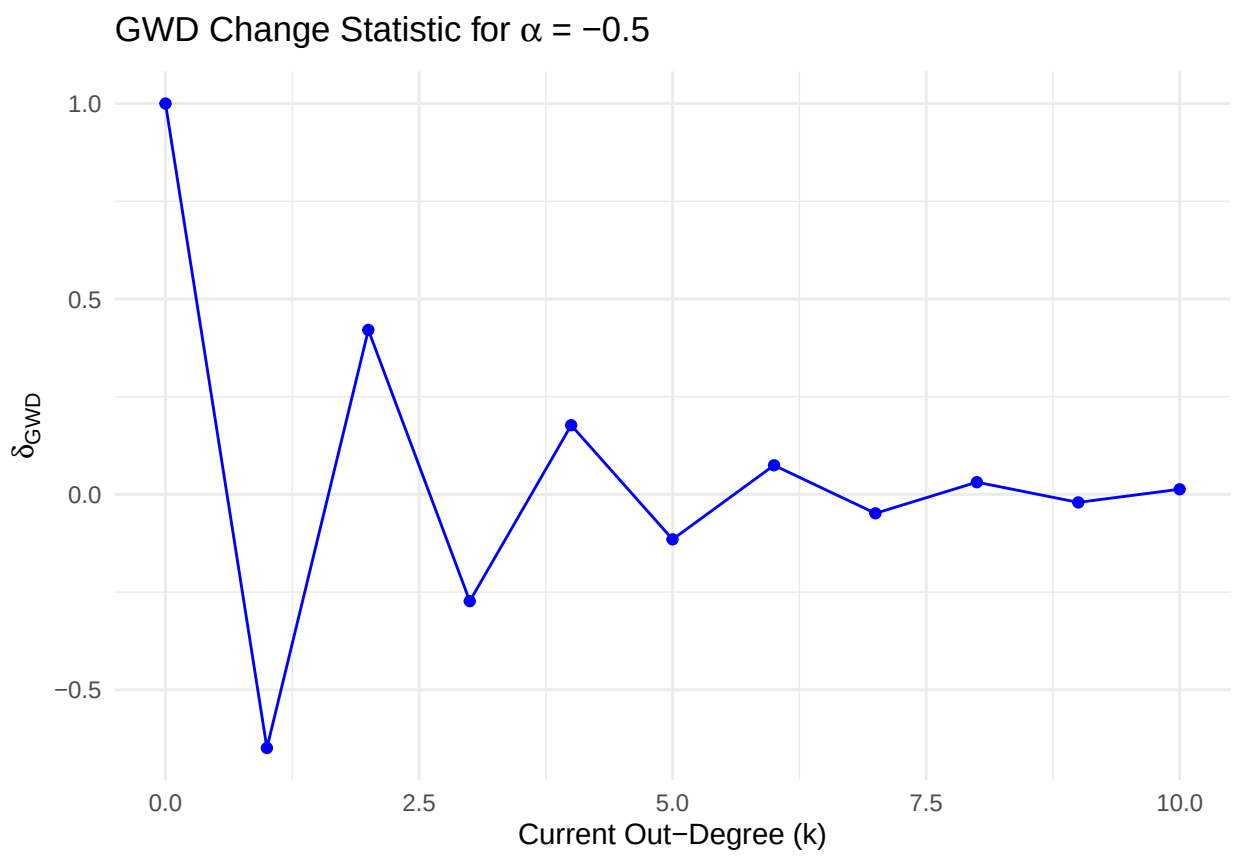
(1)

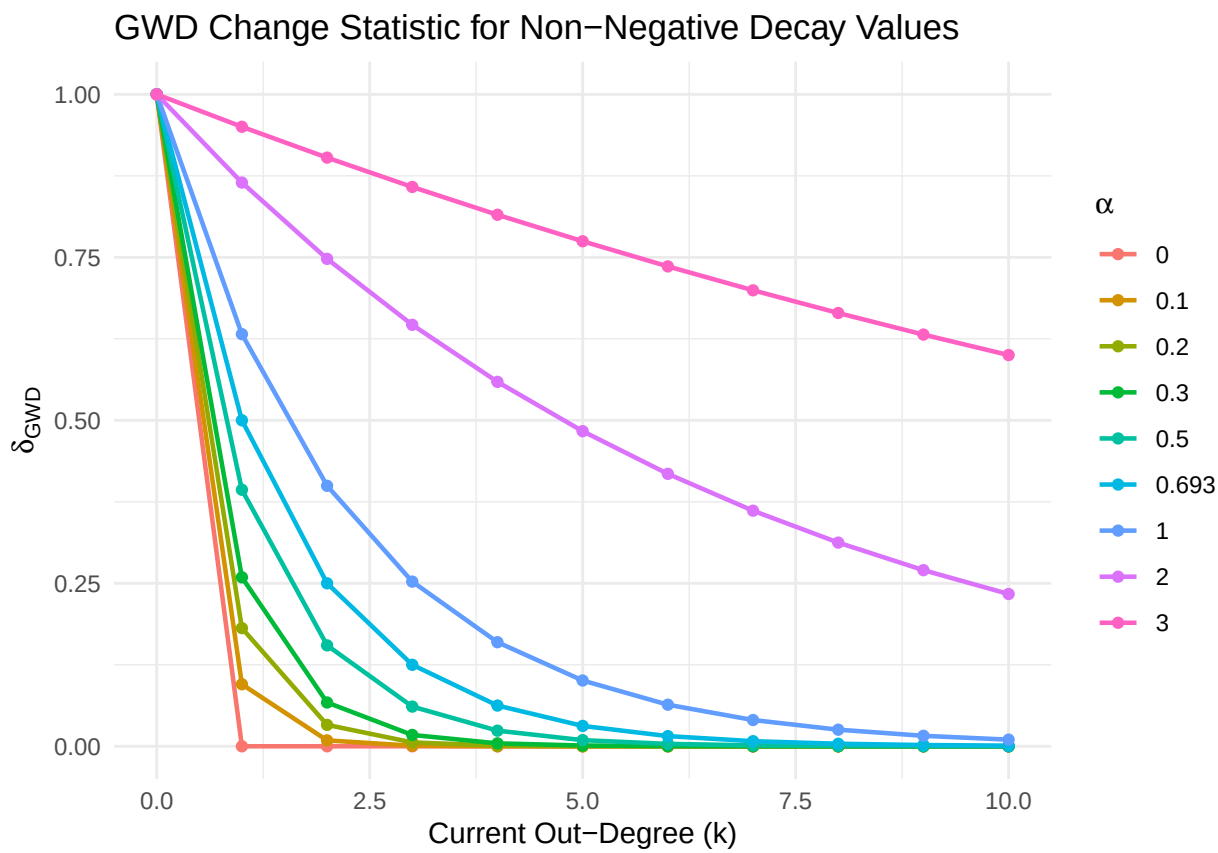
i

The difference $(x_{\text{after}} - x_{\text{before}})$ is $\delta_{\text{GWD}}(x, \alpha) = (1 - e^{-\alpha})^k$.

ii


GWD Change Statistic for α = −1
200
100
0
−100
δGWD
0.0
2.5
5.0
7.5
10.0
Current Out−Degree (k)


GWD Change Statistic for α = −0.5
1.0
0.5
0.0
−0.5
δGWD
0.0
2.5
5.0
7.5
10.0
Current Out−Degree (k)



iii

- $\alpha = -1$ - The marginal effect is close to zero and then it explodes in absolute value (oscillating between positive and negative). For popular people, the effect is massive (in absolute value).
- $\alpha = -0.5$ - The marginal effect converges to zero oscillating between positive and negative. For popular people, the effect is very small.
- $\alpha \geq 0$ - The marginal effect is positive and goes to zero as k grows. For popular people, the marginal effect is positive, but it decays quickly to zero. The greater the alpha, the slower the decay is.

iv

We would like to use a value for α that is high (maybe $\alpha \geq 0.5$), so that the marginal effect for already active people is non-negligible.

(2)

i

```
model1 <- ergm(glasgow_net1 ~ edges)
```

```
## Call:
```

```
## ergm(formula = glasgow_net1 ~ edges)
##
## Maximum Likelihood Results:
##
##       Estimate Std. Error MCMC % z value Pr(>|z|)
## edges  -3.5795     0.0479      0  -74.73   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
##  Residual Deviance:  4116  on 16511  degrees of freedom
##
## AIC: 4118  BIC: 4125  (Smaller is better. MC Std. Err. = 0)
```

ii

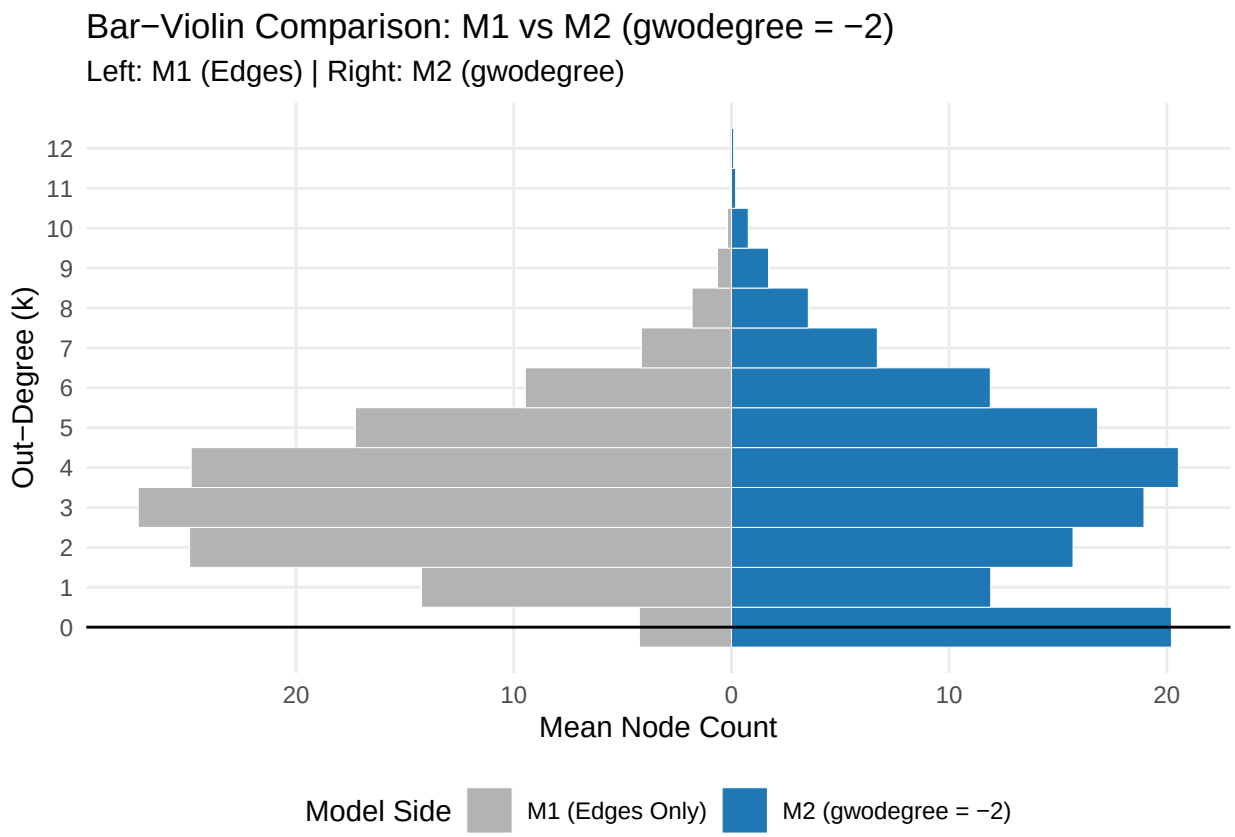
$\theta = -2$

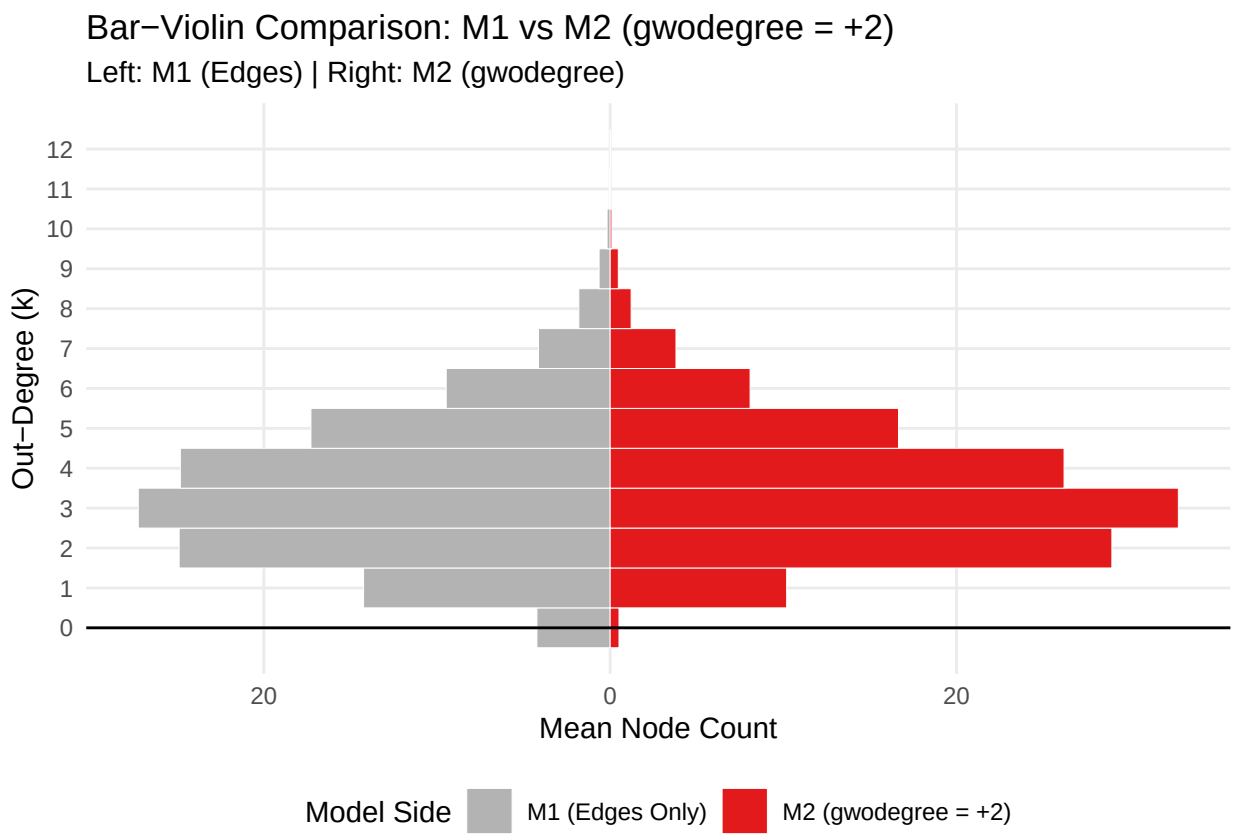
```r
model2_neg2 <- ergm(glasgow_net1 ~ edges + offset(gwodegree(0.3, fixed = TRUE)),
                    offset.coef = -2)
```

$\theta = +2$

```r
model2_neg2 <- ergm(glasgow_net1 ~ edges + offset(gwodegree(0.3, fixed = TRUE)),
                    offset.coef = +2)
```

iii


Bar−Violin Comparison: M1 vs M2 (gwodegree = −2)
Left: M1 (Edges) | Right: M2 (gwodegree)
Out−Degree (k)
Mean Node Count
Model Side
M1 (Edges Only)
M2 (gwodegree = −2)


Bar−Violin Comparison: M1 vs M2 (gwodegree = +2)
Left: M1 (Edges) | Right: M2 (gwodegree)
Out−Degree (k)
Mean Node Count
Model Side
M1 (Edges Only)
M2 (gwodegree = +2)

iv

The random network in M1 has an out-degree distribution that looks a lot like a normal distribution, which is expected as we only control for the number of edges.

$\theta_{\mathbf{GWD}} = -2$

Here the distribution is bimodal - out-degree values that are either 0 (isolates) or "average" are more common than the other ones. The second mode is a bit higher than the mode in M1, since the expected density is fixed and these nodes "compensate" for the larger amount of isolates.

$\theta_{\mathbf{GWD}} = +2$

Here the distribution looks like a normal distribution, but much more narrow. That is because the network is rewarder for having higher degree values. The reason that the mode is not at a very high out-degree is that there are diminishing returns at $\alpha = 0.3$, so having a more "balanced allocation" is more preferable.